DATA ENGINEERING ASSIGNMENT - 02

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Q1. What are the key stages of data life cycle, and how do they differ from the data engineering life cycle.

Ans:- INTRODUCTION

The data lifecycle represents the complete lifespan of data from creation to storage and usage. The data engineering lifecycle is a subset of the data lifecycle & focuses only on the stages that are directly controlled by data engineers.

KEY STAGES OF DATA LIFE CYCLE :

1. Data Generation (source systems) -

Data is generated from source systems such as application databases, IOT devices, and message queues. Data engineers do not own these systems but must understand how data is produced, its rate, quality, and schema.

key considerations :

- Type of source systems.
- Rate and volume of data generation.
- Data quality, duplicates, and late-arriving data.
- Schema structure and schema changes.

2. Storage -

Storage is required to hold data for processing and analysis. Modern architectures use multiple storage systems such as data warehouses and object storage. Storage decisions impact performance, scalability, and data usage across the

Key Considerations :

- Read / write performance
- Scalability.
- Schema enforcement or flexibility
- Metadata and data governance

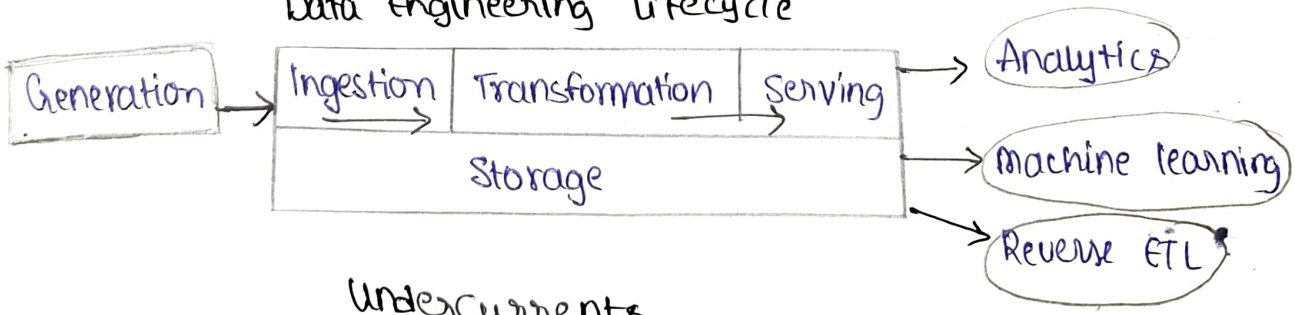
3. Data Ingestion :

Data Ingestion involves collecting data from source systems into storage. This stage is often a bottleneck because source systems maybe unreliable or outside the data engineer's control. Ingestion failures can disrupt the entire data pipeline.

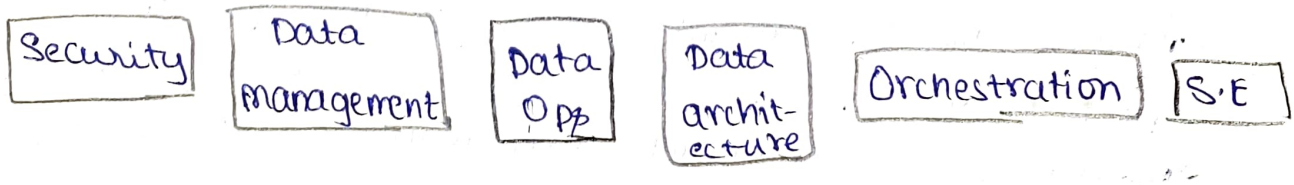
DIFFERENCE BETWEEN DATA LIFECYCLE & DATA ENGINEERING LIFECYCLE

- The data lifecycle covers data from creation to its entire lifespan.
- The data engineering lifecycle focuses only on stages managed by data engineers.
- The data engineering lifecycle is a subset of the overall data lifecycle

Data Engineering Lifecycle



Undercurrents



7. Explain the most popular uses of data - data analytics, ML and reverse ETL in detail.

Ans:- INTRODUCTION

Data is most commonly used for data analytics, machine learning (ML), and reverse ETL. These uses represent how organizations extract value from data at different levels of data maturity.

1. Data Analytics -

Data analytics is the foundation of data usage in organizations. It involves analyzing historical and current data to generate insights for decision-making. Most companies are advised to build strong analytics capabilities before moving to advanced use cases.

Key Points:

- Helps understand business performance & trends
- Uses data warehouses & analytics pipelines.
- Acts as a prerequisite for ML adoption.

2. Machine Learning (ML)

ML is one of the most impactful uses of data & typically emerges once an organization reaches a high level of data maturity. ML enables systems to learn patterns from data & make predictions or decisions automatically.

Data engineers often support ML by managing data pipelines, spark clusters, orchestration systems, and metadata tools.

Important ML considerations:

- Data quality for feature engineering.

- Data discoverability for ML teams
- Clear boundaries between data engineering & ML engineering
- Avoiding bias & ensuring ground truth.

3. Reverse ETL -

Reverse ETL refers to the process of sending processed data from analytics systems back into source systems. Earlier, this was seen as an antipattern, but it is now recognized as both useful & necessary.

Key Points :

- Feeds analytics results back into operational systems.
- Common in SaaS and advertising platforms.
- Overlaps with data engineering & ML engineering
- Expected to grow in importance.

Q3. Define DataOps. Discuss in detail the three pillars of DataOps and their purpose.

Ans:- DataOps :

DataOps is a set of practices that applies concepts from Agile methodology, DevOps, and statistical Process Control (SPC) to data systems. Just as DevOps improves the quality and speed of software delivery, DataOps improves the delivery, reliability, and quality of data products.

Unlike software products, data products are built around business logic, metrics, and decision-making, so data engineers must understand both

technical systems and business requirements.

DataOps combines people, processes, and technology to reduce time to value, promote collaboration, ensure high data quality, and support rapid experimentation. It is primarily a cultural approach, emphasizing communication, continuous learning, and iteration, supported by appropriate tools & automation.

* 3 PILLARS OF DATAOPS

1) Automation - Automation ensures consistency & reliability in data workflows. It includes version control, CI/CD, configuration as code, and orchestration tools like Airflow.

Purpose: Reduce manual errors, manage complex pipelines, and enable faster.

2.) Monitoring and Observability - Monitoring & Observability provide visibility into data pipelines, system health, and data quality. They help detect issues such as pipelines failures, bad data, and data or model drift.

Purpose: Identify problems early, prevent silent data errors, and maintain trust in data systems.

3.) Incident Response - Incident response focuses on quickly detecting, diagnosing, and resolving data failures using automation and observability. It also emphasizes blameless communication & proactive issue handling.